

DSP Lab-6

Correlation of Discrete Time Signals

1. Correlation in DSP

A mathematical operation that closely resembles convolution is correlation. Just as in the case of convolution, two signal sequences are involved in correlation. In contrast to convolution, however, our objective in computing the correlation between the two signals is to measure the degree to which the two signals are similar, and thus to extract some information that depends to a large extent on the application. Correlation of signals is often encountered in radar, sonar, digital communications, geology, and other areas in science and engineering.

Correlation is the measure of similarity or degree of similarity between two signals or a signal with itself. The general formula for correlation is given as:

$$r_{x_1x_2}[m] = \sum_{n=-\infty}^{\infty} x_1[n] x_2[n - m]$$

2. Types of Correlation

2.1 Auto correlation

Suppose we have a discrete-time signal $x[n]$. The **autocorrelation** of this signal, represented as

$$r_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n)x(n-l)$$

measures how similar the signal is to a shifted version of itself. In other words, we multiply the signal $x[n]$ with its delayed version $x[n-l]$ and sum over all n . The variable l (called **lag** or **shift**) represents how much the signal is shifted in time. If the signal and its shifted version are very similar, the autocorrelation value will be large for that lag. Autocorrelation is often used to find repeating patterns or periodicity in signals, such as identifying frequencies in a waveform or detecting time delays.

2.2 Cross correlation

Suppose we have two discrete-time signals $x[n]$ and $y[n]$. The cross-correlation between them is given by

$$r_{yx}(l) = \sum_{n=-\infty}^{\infty} y(n)x(n-l)$$

This operation measures how similar the two signals are as one signal is shifted in time relative to the other. The variable l represents the amount of shift (or lag). When $r_{yx}(l)$ reaches its maximum value, it indicates the point where the two signals are most closely aligned. Cross-correlation is widely used in signal processing applications such as pattern recognition, signal detection, and finding time delays between two related signals.

3. MATLAB Program

3.1 Using xcorr

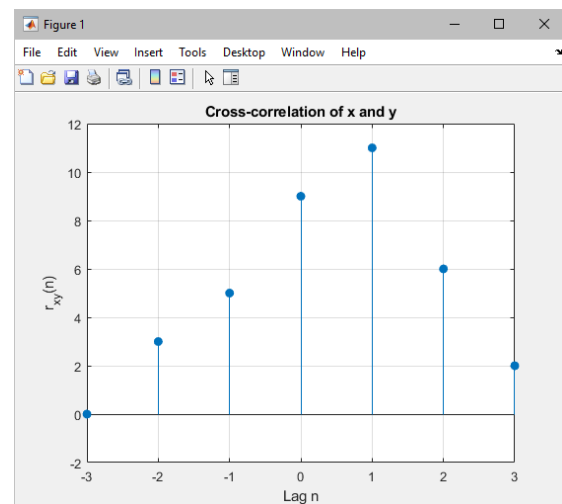
In MATLAB, the function ***xcorr(x,y)*** is used to compute the cross-correlation between two discrete-time signals x and y . It shows how similar the signals are at different time shifts (lags). If the signals have different lengths, MATLAB automatically pads zeros to match their sizes before calculating the correlation.

```
x = [1, 1, 2, 2];
y = [1, 2, 3];

if length(x) > length(y)
    y = [y, zeros(1, length(x) - length(y))]; % pad y with zeros
elseif length(y) > length(x)
    x = [x, zeros(1, length(y) - length(x))]; % pad x with zeros
end
r = xcorr(x, y);

% Creating shift vector
n = -(length(y)-1):(length(x)-1)

stem(n, r, 'filled');
xlabel('Lag n');
ylabel('r_{xy}(n)');
title('Cross-correlation of x and y');
grid on;
```



3.2 Through loop

```

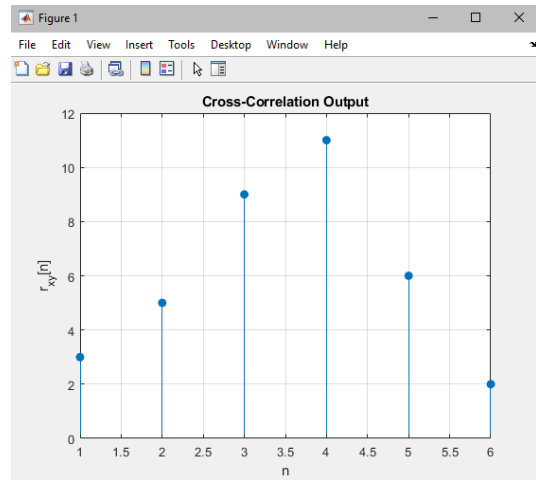
x = [1, 1, 2, 2];
h = [1, 2, 3];

xlen = length(x);
hlen = length(h);

% Initialize output matrix
y = zeros(1, xlen + hlen - 1)

% Cross-correlation computation
for i = 1:xlen
    for j = 1:hlen
        y(i + j - 1) = y(i + j - 1) + x(i) * h(hlen - j + 1);
    end
end
z = sum(y);
disp('Output (sum of correlation values):');
disp(z);
disp('Output sequence:');
disp(y);
stem(y, 'filled');

```



In Lab Task

Inlab Task 01: Determine the cross-correlation sequence $r_{xy}(l)$ of the sequence through $rcorr()$ and loop.

$$x(n) = \{\dots, 0, 0, 2, -1, 3, 7, \underset{\uparrow}{1}, 2, -3, 0, 0, \dots\}$$

$$y(n) = \{\dots, 0, 0, 1, -1, 2, -2, \underset{\uparrow}{4}, 1, -2, 5, 0, 0, \dots\}$$

Inlab Task 02: Determine the auto-correlation of the following sequence through $rcorr()$ and loops.

(a) $x(n) = \{1, 2, \underset{\uparrow}{1}, 1\}$

(b) $y(n) = \{1, 1, \underset{\uparrow}{2}, 1\}$

Post Lab Task

Postlab Task 01: Determine the normalized autocorrelation sequence of the signal $x(n)$ given by

$$x(n) = \begin{cases} 1, & -N \leq n \leq N \\ 0, & \text{otherwise} \end{cases}$$